

List of Presentations and Talks of Oliver Gutsche

Oliver Gutsche
May 10, 2026

1. Scientific Computing and Scientific Software Infrastructure - High Performance Computing (HPC)

April 2025 — Presentation for the [Chicagoland Computational Traineeship in High Energy Particle Physics](#), [\(Material\)](#)

2. Scientific Computing and Scientific Software Infrastructure - Particle Physics Overview

April 2025 — Presentation for the [Chicagoland Computational Traineeship in High Energy Particle Physics](#), [\(Material\)](#)

3. Computing and Software Infrastructure

July 2024 — Presentation at the [Fermilab-CERN HCP Summer School 2024](#), [\(Material\)](#)

4. HL-LHC Computing

July 2023 — Presentation at the [USCMS Undergraduate Summer Internship 2023](#), [\(Material\)](#)

5. The U.S. CMS HL-LHC R&D Strategic Plan

May 2023 — Parallel Session Talk at the [26th International Conference on Computing in High Energy & Nuclear Physics \(CHEP2023\)](#), [\(Material\)](#)

6. Computing

September 2022 — Plenary talk given at the DOE/HEP Review of the Energy Frontier Laboratory Research Program, [\(Material available upon request\)](#)

7. Computing

August 2022 — Lecture given at [17th Hadron Collider Physics Summer School](#), [\(Material\)](#)

8. Exabytes and the Large Hadron Collider: the importance of computing excellence for scientific discovery

November 2019 — Seminar given at [Dr. Leonard Weisenthal Colloquium Series at Lewis University](#), [\(Material\)](#)

9. WLCG Web Proxy Auto Discovery for Dynamically Created Web Proxies

November 2019 — Parallel Session Talk at the [24th International Conference on Computing in High Energy & Nuclear Physics \(CHEP2019\)](#), [\(Material\)](#)

10. Bringing heterogeneity to the CMS software framework

November 2019 — Parallel Session Talk at the [24th International Conference on Computing in High Energy & Nuclear Physics \(CHEP2019\)](#), [\(Material\)](#)

11. Highlights from Parallel Session: Software & Computing

June 2019 — Plenary Talk at the [2019 US CMS Annual Collaboration Meeting](#), [\(Material\)](#)

12. Software & Computing

June 2019 — Plenary Talk at the [2019 US CMS Annual Collaboration Meeting](#), [\(Material\)](#)

13. CMS Software, Computing and Analysis Tools

January 2019 — Plenary Talk at the CMS Data Analysis School 2019 [\(CMSDAS 2019\)](#), [\(Material\)](#)

14. Exploring the Energy Frontier using Amazon Web Services

November 2018 — Amazon Booth at [Supercomputing 2018](#) in Dallas, TX, [\(Material\)](#)

15. The Future of Large Scale Scientific Computing - Trends in computing hardware

July 2018 — [US ATLAS Workshop 2018 Open Day](#) at the University of Pittsburgh, ([Material](#))

16. The Future of Large Scale Scientific Computing

April 2018 — [Colloquium](#) given at the Physics Department at the University of Massachusetts Amherst, ([Material](#))

17. CMS Software, Computing and Analysis Tools

January 2018 — Plenary Talk at the CMS Data Analysis School 2018 ([CMSDAS 2018](#)), ([Material](#))

18. Data Organisation, Management and Access (DOMA) in High Energy Physics

November 2017 — Talk at the Workshop for Data Organisation, Management and Access (DOMA) in Astronomy, Genomics and High Energy Physics ([DOMA2017](#)), ([Material](#))

19. The Future of Large Scale Scientific Computing

October 2017 — [Colloquium](#) given at the Department of Physics and Astronomy at Texas Tech University, ([Material](#))

20. Particle Physics - A world wide journey from recording particles to analysis using big computing

September 2017 — Plenary Talk at the Chicago Council on Science & Technology Panel: [Fermilab and the New Frontiers of Physics](#), ([Material](#))

21. CMS Analysis and Data Reduction with Apache Spark

August 2017 — Parallel Session Talk at the 18th International Workshop on Advanced Computing and Analysis Techniques in Physics Research ([ACAT 2017](#)), ([Material](#))

22. 50 years Fermilab - Computing Innovations

June 2017 — Talk at the Fermilab 50th Anniversary [Symposium](#), ([Material](#))

23. Status and Plans of the CMS Big Data Project

May 2017 — Talk at the [CERN Database Futures Workshop 2017](#), ([Material](#))

24. Data Management, Access and Organisation (DOMA)

May 2017 — Summary Talk at the [2nd S2I2 HEP/CS Workshop](#), ([Material](#))

25. Data Analytics in Physics Data Reduction

April 2017 — Talk at the [CERN openlab Workshop on Machine Learning and Data Analytics 2017](#), ([Material](#))

26. Panel Discussion on the S2I2 Conceptualization for HL-LHC

March 2017 — Talk at the Openscience Grid All Hands Meeting ([OSG AHM 2017](#)), ([Material](#))

27. The Fermilab HEPCloud, or How to add 240 TFlops in an Hour or Two

January 2017 — Talk at HEP Software Foundation [Workshop](#), ([Material](#))

28. Intel Big Data Project: CMS Physics Data Reduction Use Case

December 2016 — Talk at the [CERN openlab Technical Workshop 2016](#), ([Material](#))

29. Big Data in HEP: a Comprehensive Use Case Study

October 2016 — Parallel Session Talk at 22nd International Conference on Computing in High Energy and Nuclear Physics ([CHEP 2016](#)), ([Material](#))

30. Big Data at FNAL

October 2016 — Plenary Talk given at the [8th INFIERI Workshop](#), ([Material](#))

31. Amazon and the Future of Computing

October 2016 — [Ask-A-Scientist](#) Lecture given to the general public at Fermi National Accelerator Laboratory, ([Material](#))

32. Exascale and Exabytes: Future directions in HEP software and computing

November 2015 — Seminar given at [Center of Research Computing Seminar Series](#) the University of Notre Dame, [\(Material\)](#)

33. Exascale and Exabytes - Computing at Fermilab for LHC and Intensity Frontier

August 2015 — Seminar given at the University Los Andes in Bogota, Colombia, [\(Material\)](#)

34. Future directions in HEP Software and Computing

August 2015 — Plenary talk given at the Meeting of the Division of Particles & Fields of the American Physical Society at the University of Michigan [\(DPF2015\)](#), [\(Material\)](#)

35. Diversity in Computing Technologies and Strategies for Dynamic Resource Allocation

April 2015 — Plenary talk given at the Conference for Computing in High Energy Physics [\(CHEP2015\)](#), Okinawa, Japan , [\(Material\)](#)

36. CMS Computing Operations: Outlook on LHC Run 2

March 2015 — Talk given at the Openscience Grid All Hands Meeting 2015 at Northwestern University [\(OSGAHM2015\)](#), [\(Material\)](#)

37. Computing Facilities and Services

February 2015 — Talk given at the [Fermilab Institutional Review 2015](#), [\(Material\)](#)

38. Scientific Computing

June 2014 — Plenary talk given at the [47th Annual Fermilab Users Meeting](#), [\(Material\)](#)

39. CMS computing operations: the year ahead

April 2014 — Talk given at the Openscience Grid All Hands Meeting 2014 at SLAC [\(OSGAHM2014\)](#), [\(Material\)](#)

40. Recent Top Pair Asymmetry Measurements at CMS

February 2014 — Seminar given at the [Fermilab Joint Experimental-Theoretical Physics Seminar](#), [\(Material\)](#)

41. Computing: From Data to Physics

December 2013 — [Ask-A-Scientist](#) Lecture given to the general public at Fermi National Accelerator Laboratory, [\(Material\)](#)

42. Measurement of top quark properties with CMS

November 2013 — Parallel session talk given at the 19th International Symposium on Particles, Strings, and Cosmology [\(PASCOS2013\)](#), Taipei, Taiwan, [\(Material\)](#)

43. CMS: T2 dynamic data placement and plans for T2 disk space handling

November 2013 — Plenary talk given at the [2013 WLCG Collaboration Workshop](#) in Copenhagen, Denmark, [\(Material\)](#)

44. CMS: Disk/Tape Separation

November 2013 — Plenary talk given at the [2013 WLCG Collaboration Workshop](#) in Copenhagen, Denmark, [\(Material\)](#)

45. Computing Support Facilities

November 2013 — Talk given at the [2013 Fermi National Accelerator Laboratory Annual Science and Technology Review](#), [\(Material\)](#)

46. CMS computing operations in LHC run 1

October 2013 — Parallel session talk given at the 2013 Conference for Computing in High Energy Physics [\(CHEP2013\)](#), Amsterdam, The Netherlands, [\(Material\)](#)

47. CMS Production Transfer Case Study

August 2013 — Talk given at the [2013 Department of Energy ESNet Requirements Review](#), [\(Material\)](#)

48. Summary of CMS Operations Procedures

May 2012 — Talk given at the [WLCG Collaboration Workshop 2012](#), ([Material](#))

49. Summary of Event Processing Track

October 2010 — Plenary talk given at the conference for Computing in High Energy and Nuclear Physics 2010 ([CHEP2010](#)) in Taipei (Taiwan), ([Material](#))

50. Validation of software releases for CMS

March 2009 — Parallel session talk given at the conference for Computing in High Energy and Nuclear Physics 2009 ([CHEP2009](#)) in Prague, Czech Republic, ([Material](#))

51. Dilepton Final States Studies at CMS: From the Standard Model to New Physics

January 2009 — Talk given at the [LHC and Dark Matter Workshop](#) at the University of Michigan in Ann Arbor, MI, ([Material](#))

52. WLCG scale testing during CMS data challenges

September 2007 — Parallel session talk given at the conference for Computing in High Energy and Nuclear Physics 2007 ([CHEP2007](#)) in Victoria, Canada, ([Material](#))

53. Distributed CMS Analysis on the Open Science Grid

February 2006 — Parallel session talk given at the conference for Computing in High Energy and Nuclear Physics 2006 ([CHEP2006](#)) in Mumbai, India, ([Material](#))